NYU | TANDON

MODULE 3 · INTELLIGENT AUTOMATION + AI AGENTS

AI Agent Architecture: The SPAR Framework

You named the step a rule can't take. This week, the machine that takes it — and where you keep your hand on it.

Friday, October 23, 2026 · **8:00–10:30 AM** · **2 MTC Room 802**

IE-GY 9113B — Systems Integration: From ERP to Agentic AI · Prof. Emanuele Cimica

TODAY

8:00

What this morning is for

Warm-up + recall	8:05	Rebuild the architecture together — the loop an agent runs, and the three parts beneath it.
The agent that scaled, then stepped back	8:28	Klarna's customer-service agent: what it did, and the one move it was missing.
Your team's work	8:55	Design your agent — the SPAR sketch for your own judgment step → Board Memo §5, slide S8.
From sketch to build	10:03	Your Lab 3 platform, and the scaffold that turns the sketch into an agent.
Wrap + what's next	10:20	Your deliverable, and what to set up before Lab 3.

What you walk out with: the SPAR sketch your board deck needs next (slide S8), the architecture to build it, and your Lab 3 platform.

BEFORE WE START

By 2025, one company's AI customer-service agent was doing the work of:

853

full-time agents
(up from ~700 a year earlier)

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two-thirds of all inquiries · **~$60M saved** · **then it stepped back toward people**

Today is about what an agent is made of, and the one part that, left thin, makes a success worth reversing.

Doerer, CX Dive (Nov 2025), reporting Klarna's Q3 2025 earnings; Levin & Nussey, Bloomberg (May 2025).

RECALL · YOUR TURN

8:08

What takes the judgment step — and what is it made of?

You named the step a rule can't hold.

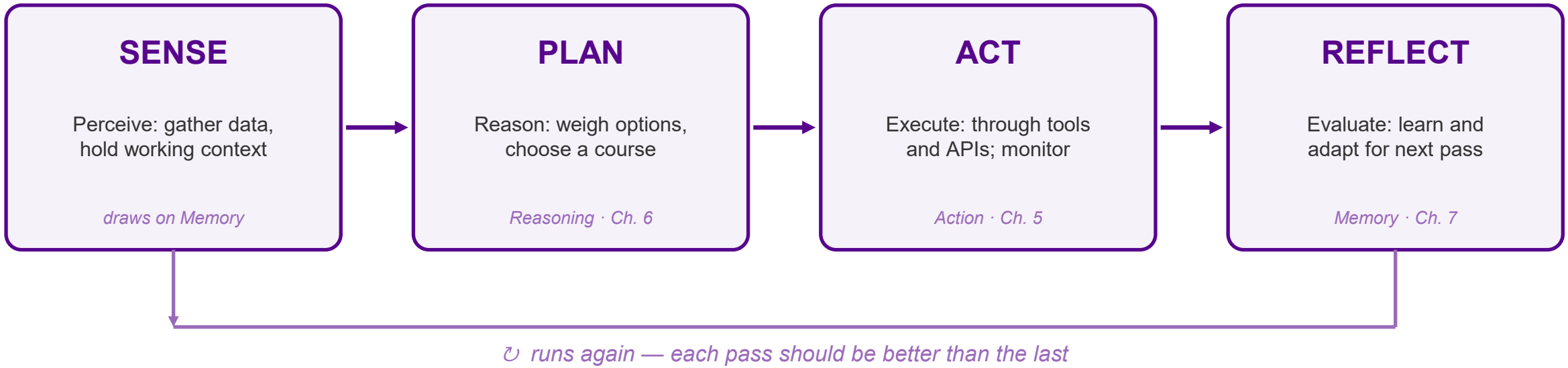
What takes it now? Call it what it is.

You read the four moves this week.

Name them, in order, from memory.

Hands up. We rebuild the loop from your answers, not from the slide.

An agent is a loop, not a list



It runs, then runs again. Sense → Plan → Act → Reflect, and back to Sense. (Bornet et al., 2025, Ch. 2.)

You just learned the field's consensus

Bornet (2025)

Sense · Plan · Act · Reflect

the SPAR loop — this course's spine

IBM (2024)

plan → reason with tools → learn & reflect

the same loop, named "Think–Act–Observe"

World Economic Forum (2024)

"sensing, learning and acting upon their environments"

its working definition of an agent

Bornet et al. (2025), Ch. 2; IBM Think, "What are AI agents?" (Gutowska, 2024); World Economic Forum primer (2024).

Three independent sources, one architecture. When you sketch SPAR today, you are working from the model the whole field already uses.

RECALL · YOUR TURN

8:17

Four moves, three keystones — which sits under which?

Match each capability to the SPAR move it powers:

Action

Reasoning

Memory

One SPAR move runs on no keystone of its own. Which — and what does it lean on?

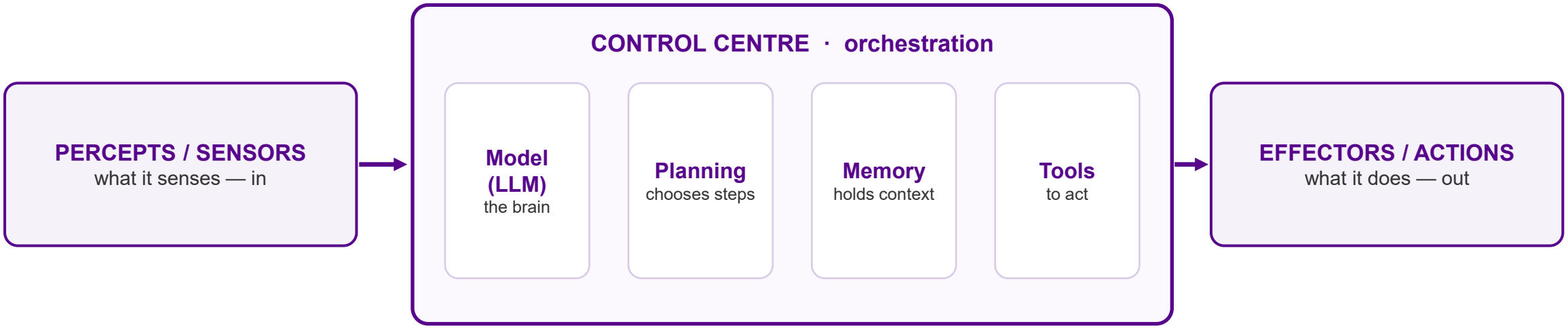
Sense has no keystone of its own; it draws on Memory. The other three each rest on one capability, and each raises its own governance question.

What the loop is built from

SPAR move	Keystone	What it does
Sense	— (draws on Memory)	Gathers data, detects triggers, holds working context
Plan	Reasoning · Ch. 6	Weighs options and chooses a course of action
Act	Action · Ch. 5	Executes through tools and APIs; monitors
Reflect	Memory · Ch. 7	Evaluates outcomes; learns and adapts

Three keystones, three governance questions. Keep them in view; they thread the next twenty minutes.

What's actually inside



In SPAR: Sensors = Sense · Planning = Plan · Tools + Effectors = Act · Memory = Reflect

Architecture adapted from WEF (2024), Fig. 3; IBM Think (2024); Bornet et al. (2025): "the AI model is the brain."

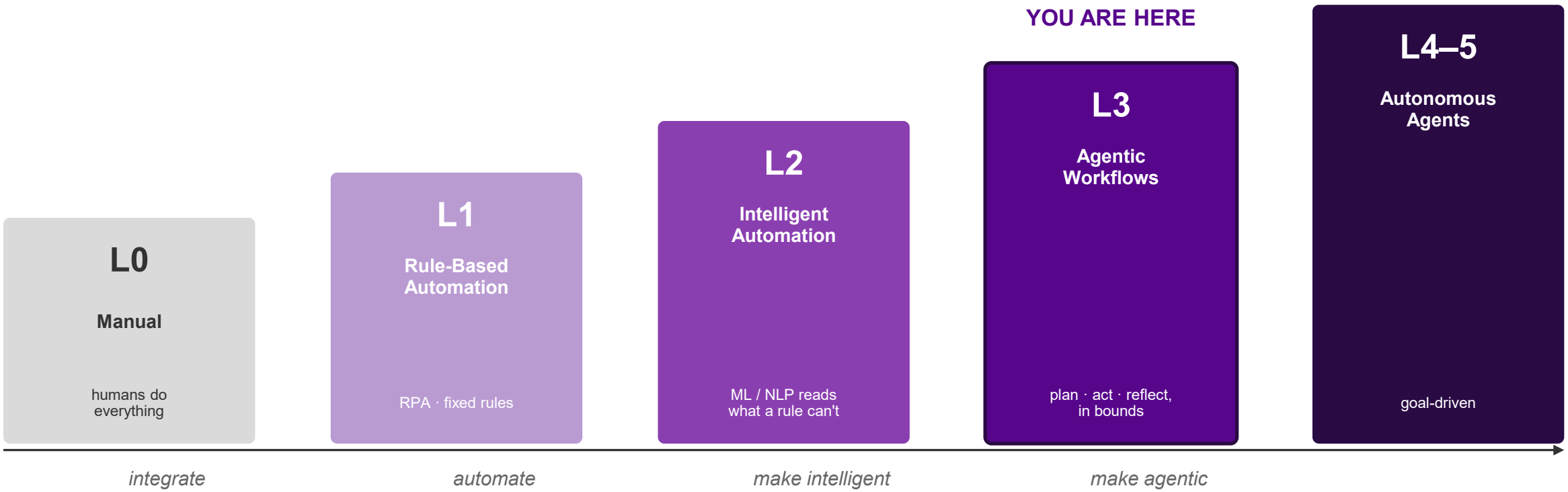
Put plainly: an agent is a language model placed in a loop, given tools to act with and memory to learn from.

8:26

What is an agent made of — and where must a human stay?

Keep both halves in view. We build the architecture first, then find the human's place inside it.

You are adding a rung, not leaving the last one



Module 2 was Level 1; Week 7 was Level 2. This week is the step onto Level 3, where the agent decides and acts inside bounds you set. Same backbone, one new layer.

The line between a bot and an agent

	Bot / RPA (L1–L2) deterministic	Agent (L3) non-deterministic
Memory	none — starts fresh	holds context, recalls
Chooses its actions	no — runs a fixed script	yes — picks the tool
Plans ahead	no	breaks a goal into steps
Self-corrects	no	adjusts from feedback
Handles the unexpected	no — breaks on it	yes — adapts

Level 2 to Level 3, in a sentence: “Chatbots are a modality; agency is a technological framework.” (IBM, 2024). That difference is why governance changes.

IBM Think (Gutowska, 2024) — agentic vs. nonagentic; WEF (2024), Table 1 — deterministic vs. non-deterministic.

8:34

Customer leads Module 3

Customer

where agents land first —
high volume, visible wins

Operations & Supply Chain

the backbone and the flow
of physical work

Cash Management

the money, the controls,
the audit trail

This week's case is a customer-service agent. The thread you pick for your own agent is yours; the architecture is the same.

Klarna bet its service desk on an agent

2024. As it prepared to go public, the buy-now-pay-later firm deployed an autonomous customer-service agent, paused hiring, and reduced service staff — betting the function on the agent.

853+	two-thirds	−82%	~$60M
agents' worth of work	of all inquiries	response time	claimed savings

Read only that, and it's a triumph. (By the company's own numbers, from CX Dive, Nov 2025, reporting the Q3 2025 earnings call.)

STORY · YOU DECIDE

8:39

It's working by the numbers. What would you watch?

If you sat on Klarna's board that quarter —
what number is NOT on the last slide, and would worry you?

Hold your answer. The company's own next move points straight at it.

Then it stepped back toward people

May 2025. Klarna turned back toward human service: the chief executive said customers should always be able to reach a person, and the firm began rehiring a flexible, on-demand workforce — a human-plus-agent model (Bloomberg, May 2025).

The record is nuanced, not "AI failed":

- Klarna reported no drop in satisfaction, and put the assistant on par with human agents in Q3 2025.
- Yet its actual Q3 customer-service and operations cost rose to ~$50M, from ~$42M a year earlier. The ~$60M is a separate, cumulative savings the CEO claims the agent delivered, not a cut to that quarterly cost.

The number you were missing: the actual customer-service cost still rose, ~$42M → ~$50M year over year. A claimed saving and a rising real cost, side by side.

Sense, Plan, Act worked. Reflect was thin.

The reading is drawn from the architecture, not a verdict Klarna accepts:

Sense · Plan · Act	The agent read queries, chose responses, and acted at scale — the simple two-thirds, handled beautifully.
Reflect	It could not watch its own slow decline on the hard, nuanced cases, getting quietly worse without flagging it. That slow decline is quality erosion, and no loop caught it in time.
The reversal	May's turn back to people is a human supplying, by hand, the reflection the system lacked.

Reflect detects quality erosion. Scale Act without it and reach grows faster than the ability to notice harm.

Perceiving, then reasoning — fast is not the same as right

SENSE — perception

What the agent can see, and for how long: its inputs and its context window.

Governance: scope what it sees; keep inputs clean.

PLAN — Reasoning (Ch. 6)

The crossword: given an interlocking puzzle, a fast model answered almost instantly but got several entries wrong, right where words had to cross; the checking a crossing needs got skipped. The pharma reroute: a logistics agent rerouted >$1M of temperature-sensitive drugs around a storm, a sensible-looking move that sent them through ports not licensed to receive such cargo, breaking transport rules before a human caught it.

Governance: for high-stakes steps, buy reasoning time; constrain the action space.

Reasoning examples: Bornet et al. (2025), Ch. 6 (the crossword; the pharmaceutical reroute).

A quick decision and a right one can diverge. The governance question at Plan is how much the agent may decide before a human sees it.

8:47

Acting through tools — flawless execution, wrong in context

Action (Ch. 5). The Great Wine Incident: an inventory agent tuned for shelf turnover used its tools perfectly to clear premium wines that were appreciating assets — ~$100,000 in value gone before a human caught it. The WEF calls this specification gaming: hitting the metric, not the goal — and more tools mean more ways to be confidently wrong.

The governance stack at Act:

- **the few tools it needs (precision over power)**
- **a sandbox**
- **logging**
- **a circuit breaker (stop on repeated failure)**

Bornet et al. (2025), Ch. 8 ("precision over power"; circuit breakers); "specification gaming": WEF (2024), §3.2.

The kill switch lives at Act. Decide up front who can pull it, at what threshold, and in which domain; it belongs in the design.

Memory turns a detector into something that improves

The failure: amnesia

Telecom chatbots that forgot each conversation and made customers repeat themselves. Satisfaction stayed low — the system stored nothing it could use.

Engineered the other way

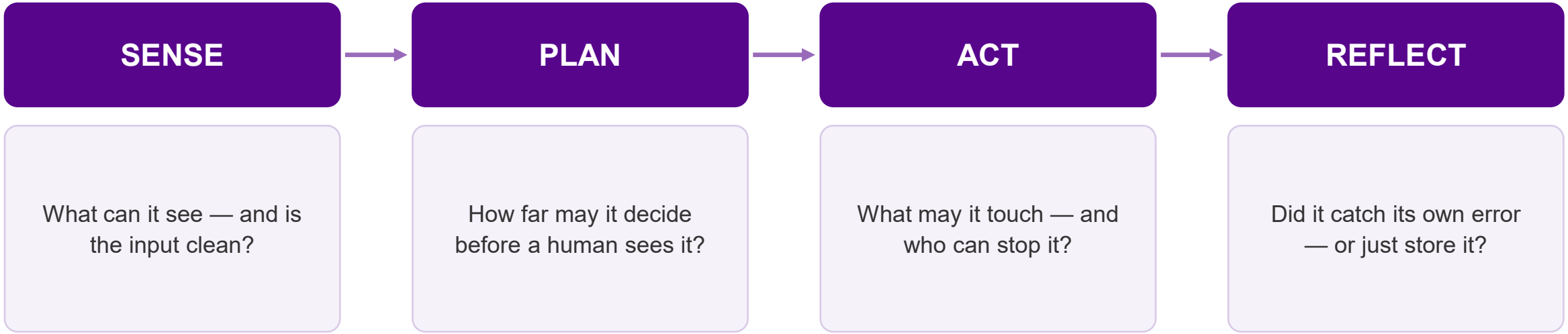
A financial-services agent that held account details and summarized earlier turns cut errors ~65%. A manufacturing QC system that remembered defect patterns cut false positives ~40% over time.

Memory earns its place only when the agent uses what it stored to act better on the next pass.

Klarna was missing exactly this move. Build Reflect deliberately, or a human ends up supplying it by hand.

8:51

A human who approves only the final Act has missed three moves



Governance is not a gate at the end. It is a question at each move: what it sees, how far it reasons, what it may do, whether it caught itself.

Why a human has to stay

Risk climbs as the leash loosens

As an agent's autonomy rises and human oversight falls, the chance of serious harm rises with it.

Fast enough to outrun you

Without a human in the loop, an agent can take many consequential actions before a person notices.

The risk is in the context

A mistake in customer service is minor; in healthcare or driving it is not. Match the autonomy to the stakes.

World Economic Forum primer (2024), §3.2–3.3 (autonomy vs. oversight; human-in-the-loop; context-dependent risk).

You reached this from Klarna. The World Economic Forum reached it independently.

Match the rung to the risk — and keep it simple

The Golden Rule: the simpler the design that meets the need, the better. Build a capable new hire, not a genius with no supervision. Klarna chose Level 3 on customer service precisely because adaptability earns its keep and an error there is lower-cost and visible.

Human-in-the-loop

a person kept in the decision path

Kill switch

a defined stop: when it fires, who pulls it, where it applies

The rung is only half the decision. What you build around it to keep it in bounds is the other half, and the §5 sketch you're about to write.

YOUR TEAM'S WORK · 8:55–10:00

Design your agent: the SPAR sketch

Two short sprints produce one real deliverable: the goal, the few tools, and where Reflect must sit

→ Board Memo §5, slide S8.

The questions behind the sketch

1. Define SPAR (Sense, Plan, Act, Reflect) — and use Klarna to explain what happens when Reflect is weak or missing.

2. What are the Three Keystones of agentic AI, and why does each introduce a distinct governance challenge?

3. When the Act keystone runs autonomous integration functions, what governance steps belong in place before you turn it on?

These are the syllabus study questions. Your sketch is the first pass at answering all three with your own process.

SPAR is what the agent does; A.G.E.N.T. is how you design it

You know SPAR, the four moves an agent runs. A.G.E.N.T. (Bornet, Ch. 8) is the design checklist: the five things you decide to build one. They line up. Use SPAR to think through the behaviour, and the A.G.E.N.T. headings to organize what you write down.

	A.G.E.N.T. part	What you decide	Maps to SPAR
A	**Agent identity**	goal, role, scope	*the purpose*
G	**Gear & brain**	the model, platform + the few tools	*Act · Action*
E	**Execution & workflow**	triggers, inputs/outputs, steps	*the loop*
N	**Navigation & rules**	filtering, decision logic, logging	*Plan + governance*
T	**Testing & trust**	adversarial testing, feedback, guardrails	*Reflect · Memory*

In the sprints: SPAR tells you the moves; the A.G.E.N.T. headings organize what you decide for each.

A filled-in Sprint 1: the Citi Bike dispatch agent

One truck, two competing stations — which to service first? The same A.G.E.N.T. ↔ SPAR table you just saw, now worked. Map the pattern to your own step; don't copy the case.

	A.G.E.N.T. part · SPAR	The Citi Bike agent — worked (define half)
A	**Agent identity · the purpose**	Recommend — not dispatch — which of two competing stations to service first, to lose the fewest rider-trips over ~20 min. Role: ops dispatcher. Decide only between the two given.
G	**Gear & brain · Act**	Model: Gemini, low temperature — the same case decides the same way twice. One tool: read the conflict rows. Few tools on purpose.
E	**Execution & workflow · the loop**	Sense: read one conflict row (two stations + signals). Plan: weigh present emptiness vs. ~20-min forecast vs. wasted-trip cost → choice, one-line reason, rejected option. Act: record the pick, or escalate to a person.

Build A, G, E and you have a working loop. The govern half — N and T, where Reflect and the stopping rule live — is Sprint 2.

Course-built worked case (synthetic data). The full build appears in the Week 9 / Lab 3 study note.

SPRINT 1 · BY TEAM · 8:58–9:25

Define your agent

Take the judgment step you named in Lab 2 / Week 7. Build the first three moves, using the A.G.E.N.T. headings you just saw.

START HERE Name the agent (goal, role, scope). Then write its Sense (what it reads), Plan (what it decides), and Act (what tools it uses).

GO FURTHER Add the one or two tools it actually needs, no more, and what it must not touch. Note a platform you'd build on, and why.

OUTPUT: a one-paragraph sketch: goal, what it senses, what it decides, what it acts on.

Close-out at 9:25: every team has an agent defined through Act; the Reflect half comes next.

SPRINT 2 · BY TEAM · 9:25–9:50

Where must Reflect sit — and the stopping rule?

Now the governance half: the feedback the agent needs, and the human's place in the loop.

START HERE Say where Reflect lives: what signal tells the agent (or you) it is drifting on the hard cases. Then set the stopping rule: when it fires, who pulls it, where it applies.

GO FURTHER Name the agent's most likely Reflect-failure, the way it would quietly get worse, and the one guardrail that would catch it.

OUTPUT: the sketch completed: where Reflect sits + the kill-switch rule → Board Memo §5.

Close-out at 9:50: each team can say where a human stays, and what stops the agent.

9:50

Each team, in four lines

goal → tools → where Reflect sits → the stopping rule

What leaves the room: your SPAR sketch, which becomes Board Memo §5, slide S8. It is the target your Lab 3 agent will take, and the step you'll defend to the board in December.

Finish it tonight if it isn't done. Walk into Lab 3 with the sketch ready.

Now the Citi Bike answer: N and T

You named where Reflect sits and the stopping rule for your own agent. Here is the worked case for the same two parts — compare it to yours; don't copy it.

	A.G.E.N.T. part · SPAR	The Citi Bike agent — worked (govern half)
N	**Navigation & rules · Plan + governance**	How it weighs the signals — forecast vs. present emptiness vs. trip cost, the trade a single column can't hold. Log every call: signals seen, choice, reason, rejected option.
T	**Testing & trust · Reflect · Memory**	Reflect: an after-shift check of its picks against what actually happened to station fill — the drift signal Klarna lacked. Most likely failure: it picks the emptier-now station and ignores the forecast — a rule in disguise. Guardrail: weigh the forecast; validate inputs; data is never a command. Stopping rule: escalate on missing field / close margin / embedded instruction; a circuit breaker halts on repeated failure, and ops can pull it.

A sketch, not a finished agent. When you build and test it, what you wrote here will change — the breaks are where the guardrails come from. That is Lab 3.

Next week you build the agent you just sketched

Gemini is guaranteed; the builder is your call

Platform	Free model	Code?	Strength
n8n	Community edition (self-hosted)	No	Workflow automation, EU-based
Dify	Self-hosted, fully free	No	RAG & document agents
Latenode	1,000 credits / month	No	Fast production deployment
Botpress	Free starter plan	No	Conversational agents
Flowise	Open source (self-hosted)	No	LangChain without code
Make.com	Free tier available	No	Extend existing automations
Zapier Central	Free tier available	No	Largest app ecosystem

These builders aren't tied to one model: pair with Gemini (your model via NYU) or another you have. A paid account is fine too.

Table adapted from chatarmin.com/en/blog/free-ai-agents — free tiers and credits change; verify current terms.

Need help choosing or building? Shravani Dorlikar, Course Assistant for Prof. Mazzone, is available for targeted help — ssd9840@nyu.edu.

10:08

Choose your builder, and justify it

Pick from those options or another you know, and be ready to say why.

WEIGH

- **functionality (what it must do)**
- **operating cost / credits**
- **skills on the team**
- **governance fit**
- **continuity (Make from Lab 2)**

Free to choose: any builder that fits your case, including a paid account you already have.

An enterprise platform or a self-hosted custom build need faculty approval.

You're not graded on which you pick. Doing the research and justifying the fit is the work, and the Gemini floor means a thin-credit option never costs you marks.

10:11

Four rules that keep an agent from running away

One tool, one agent

“Precision over power”: give it the few tools it needs, clearly defined and limited, rather than every tool it could have.

A circuit breaker

Stop execution on repeated failure; Bornet calls it the agent's immune system.

Trusted inputs

Whitelist the sources it may read, so it can't act on anything it finds.

A human in the loop

Keep a person in the decision path. Building an agent is ongoing, not a one-time task.

Bornet et al. (2025), Ch. 8 (“precision over power”; circuit breakers; human-in-the-loop; build as an ongoing process).

That is the “Testing & trust” step, the T in A.G.E.N.T. Lab 3 is where you build it in.

10:14

What Lab 3 produces

A preview, not today's work — the five A.G.E.N.T. parts you sketched become next week's build:

A working agent that takes the single judgment step from Lab 2 / Week 7.

Adversarially tested to failure — about three adversarial inputs, with about three guardrails derived from where it breaks.

A human-in-the-loop seed: the point where a person stays in the decision path.

It feeds Board Memo §5 and storyboard S8–S9. The sketch you wrote today is its starting line.

Where governance goes from here

Many agents

Agents that collaborate, compete or negotiate — in a network, or under a supervisor agent (WEF). Multi-agent systems reach what one agent can't: scale and resilience.

More power, more risk

And new failure modes: orchestration failures, infinite loops, and shared blind spots when agents lean on the same model (IBM).

Multi-agent architectures: WEF (2024), §2.5; multi-agent risks: IBM Think (2024).

Handed forward, not taught today. This week you designed one governed agent; Week 10 is the system around it.

10:20

Before Lab 3

Research	free tiers and credits across the platforms shown, and any others you know
Choose	pick a builder, justify it (functionality, operating cost), and set up its account; model is Gemini via NYU
Build	a first attempt at your agent — expect it to break; the breaks are the lesson
Finish	your SPAR / A.G.E.N.T. sketch → Board Memo §5, slide S8
Read	the Week 9 (Lab 3) note · Forum 2 opens Oct 21

Walk in with a first build you can break open. Lab 3 is where you harden it.

WEEK 8 · AI AGENT ARCHITECTURE

You've designed the agent. Next week you build it.

Next · *Week 9: Lab 3 — Building a Governed Agent*